

SULUH Ecosystem — Holistic Product Requirements Document

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SULUH Ecosystem — Holistic Product Requirements Document

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Executive Summary

The SULUH Ecosystem is a **sovereign, PDPA-compliant multi-agent AI platform for higher education research in Southeast Asia**. It consists of three composable open-source packages:

Package	Role	Status
IsRAG	Provenance-aware retrieval with trust scoring and cultural context	v0.1 alpha — 18/18 tests passing, working engine
Dewan Council	Multi-LLM deliberation with 3-stage protocol and auditable dissent	v0.1 alpha — working engine, mock mode available
SULUH AI	Orchestration platform: gateway, router, LangGraph workflows, PDPA+IRB gates, Vue 3 Workbench	Planned — package skeleton exists, zero implementation

The three packages form **one academic research programme**: IsRAG defines *what every retrieved fact looks like* (the data model), Council defines *how multiple LLMs deliberate*, and SULUH defines *how agents coordinate and execute*. Each is independently usable; together they form a sovereign, auditable, deliberation-grounded multi-agent platform with cultural and epistemological awareness built in by default.

Primary deployment target: 12-month pilot at Universiti Pendidikan Sultan Idris (UPSI), Malaysia — serving ~275 research staff and 2,400+ graduate students.

1. Project Vision & Context

1.1 The Sovereignty Gap

Higher education institutions in Malaysia and Southeast Asia face a regulatory paradox:

- Research data (participant records, ethics approvals, pre-publication manuscripts, indigenous knowledge) **cannot legally be processed by foreign cloud AI services** under PDPA 2010, GDPR, and sectoral rules like HIPAA and IRB.
- Universities **must adopt AI** to remain competitive in research productivity, reproducibility, and funding.

- The current landscape is fragmented: ad-hoc local GPU workstations, foreign cloud tools IRBs cannot approve, and a widening capability gap between well-resourced and under-resourced institutions.

SULUH resolves this paradox by providing a **sovereign, on-premise platform** where all data, model weights, and agent state remain within institutional jurisdiction.

1.2 Strategic Positioning

- **First-mover advantage:** First Malaysian university to deploy a sovereign, PDPA-compliant multi-agent AI research platform on a 1,280 GB distributed cluster.
- **National alignment:** Reference implementation for Malaysia National AI Agenda 2026–2030.
- **Regional template:** Consortium expansion model for partner institutions (UTM, UM, NTU, and other ASEAN universities).
- **Academic contribution:** Grounded in a 1,000-year-old epistemology (Islamic Isnad) applied to modern AI — contributes to the global decolonial AI movement.

1.3 Key Metrics (12-Month Target)

Metric	Target
Research-domain answer accuracy	93%
Swarm task success rate	90%
PDPA compliance score	100%
IRB gate accuracy	98%
Staff adoption rate	75%
Average response latency	< 2.5s
Anthropic cloud escalation share	< 5%
Productive hours returned/year	15,000–35,000 (8–18 FTE equivalent)

2. Current State Assessment

2.1 What Is Built & Working

IsRAG (packages/israg/) — v0.1 alpha 18/18 tests passing (pytest, 0.03 seconds). Full implementation:

- **Core engine:** IsRAGEngine class with ingest(), retrieve(), add_verification(), get_entry(), list_entries(), get_stats().

- **Data model:** KnowledgeEntry, ProvenanceStep, Verification-Record, TrustEngine, IsnadChain, CulturalContext, Retrieval-Result.
- **5-factor trust engine:** Verification level (30%), source credibility (25%), verification quality (20%), cross-references (15%), freshness (10%).
- **13 cultural frameworks:** Global, Western, East Asian, Southeast Asian, South Asian, Indonesian, Malaysian, Thai, Vietnamese, Filipino, Singaporean, Islamic, Buddhist, Indigenous.
- **Isn't yet:** Vector search (currently keyword-based), persistent storage (in-memory dict), embedding pipeline.

Dewan Council (packages/council/) — v0.1 alpha Working engine with full protocol:

- **3-stage deliberation:** (1) Independent first opinions □ (2) Cross-review matrix □ (3) Chairman synthesis with dissent.
- **7-member reference council:** Ketua Claude (Opus 4.7), Anggota GPT (5.5), Gemini (3 Flash), Kimi (K2.6), Deepseek (V4 Flash), Qwen (3.6 Plus), Nemotron (3 Super).
- **Mock mode:** Works without API key for testing the protocol.
- **OpenRouter integration:** Single API for all models.
- **Integration with IsRAG:** Council opinions can be grounded in IsRAG KnowledgeEntry objects.
- **Isn't yet:** Streaming, async deliberation, learned chairman selection.

Documentation

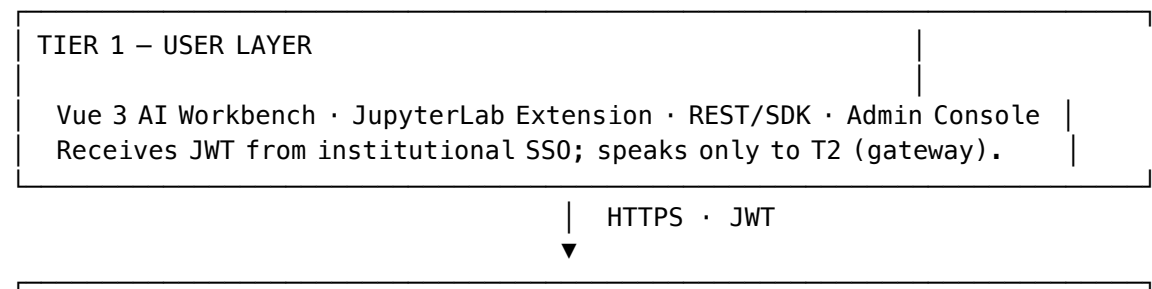
- README.md — ecosystem-level overview
- docs/ARCHITECTURE.md — 4-tier architecture with composability model (~20 KB)
- docs/PRINCIPLES.md — 7 immutable design principles, operationalised in code
- docs/COMPLIANCE.md — PDPA + IRB dual-gate model (~7 KB)
- docs/SULUH-PRD-ANALYSIS.md — analysis of the original SULUH SWARM PRD v1.0
- docs/REBRAND-KRAG-T0-ISRAG.md — migration note
- research/proposal/SULUH-ECOSYSTEM-PROPOSAL.md — unified programme proposal (RM 285K budget)
- research/papers/paper-1-israg.md — IsRAG paper draft
- research/papers/paper-2-council.md — Dewan Council paper draft
- research/papers/paper-3-suluh.md — SULUH AI paper draft

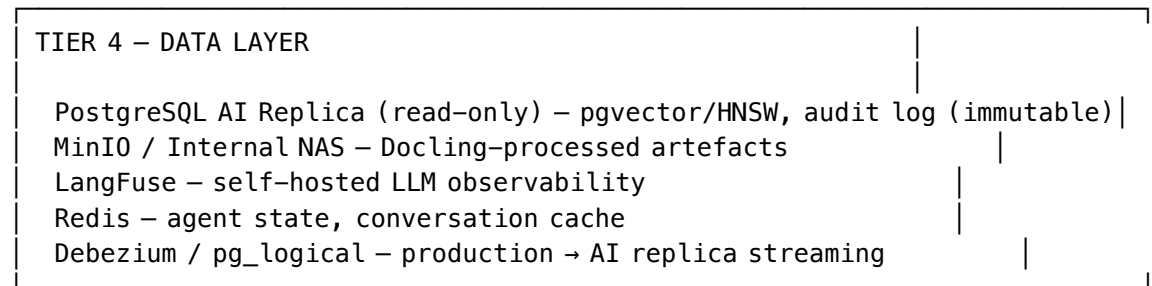
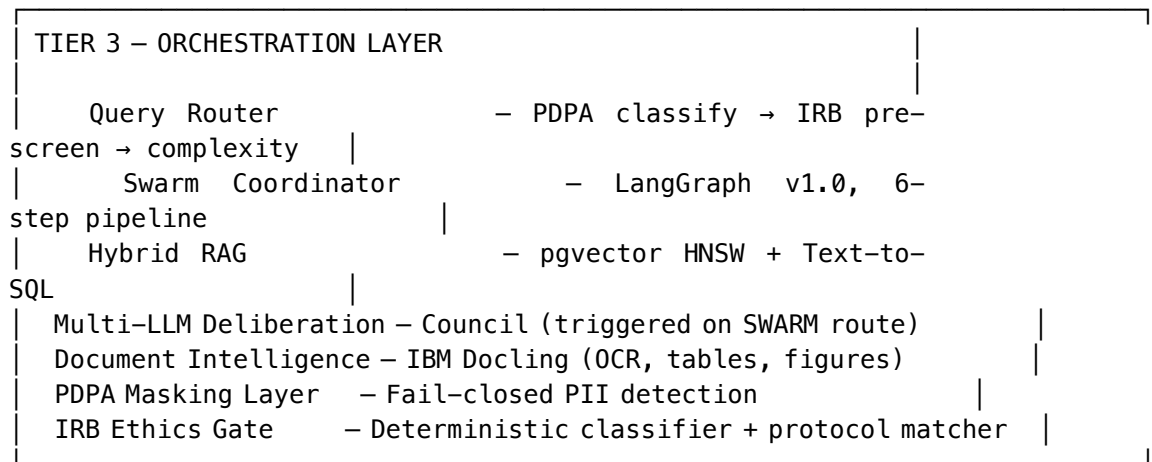
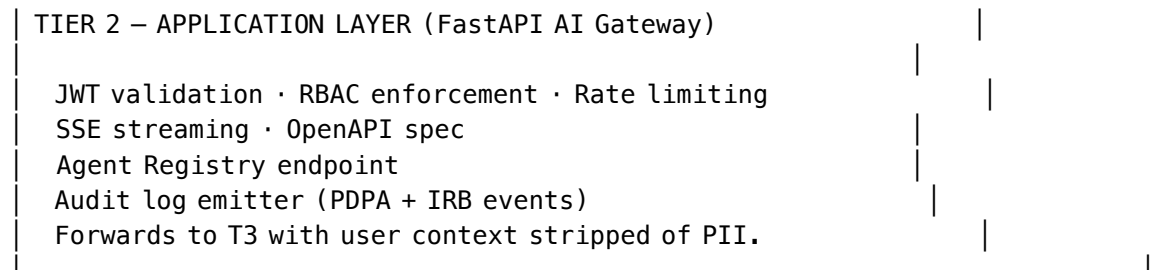
2.2 What Is NOT Built

Component	Status	Priority
SULUH FastAPI gateway (suluh/gateway/)	Zero code	P0 — Phase 0 deliverable
Query router (suluh/router/)	Zero code	P0 — Phase 0 deliverable
LangGraph Swarm Coordinator (suluh/swarm/)	Zero code	P1 — Phase 2
Hybrid RAG (suluh/rag/) — pgvector + Text-to-SQL	Zero code	P0 — Phase 1
Docling ingestion pipeline (suluh/docling/)	Zero code	P1 — Phase 1
PDPA masking layer (suluh/pdpa/)	Zero code	P0 — Phase 1
IRB ethics gate (suluh/irb/)	Zero code	P0 — Phase 0
Agent registry (suluh/registry/)	Zero code	P1 — Phase 1
Vue 3 Workbench UI	Zero code	P1 — Phase 1
Golden evaluation set	Zero entries	P0 — Phase 0
Vector search (replaces keyword matching)	Not implemented	P0 — Phase 1
Persistent storage (replaces in-memory)	Not implemented	P0 — Phase 1
Git history	No .git initialised	P0 — immediate

3. Architecture Overview

3.1 Four-Tier Design





OPTIONAL CLOUD QUALITY LAYER

Anthropic Claude API

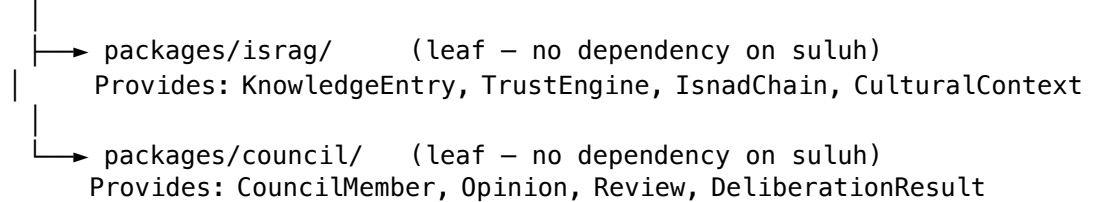
Conditions: anonymised + non-human-subject + confidence < 0.72

Target share: < 5% of all queries

Hard-blocked for: RESTRICTED data, human-subject research

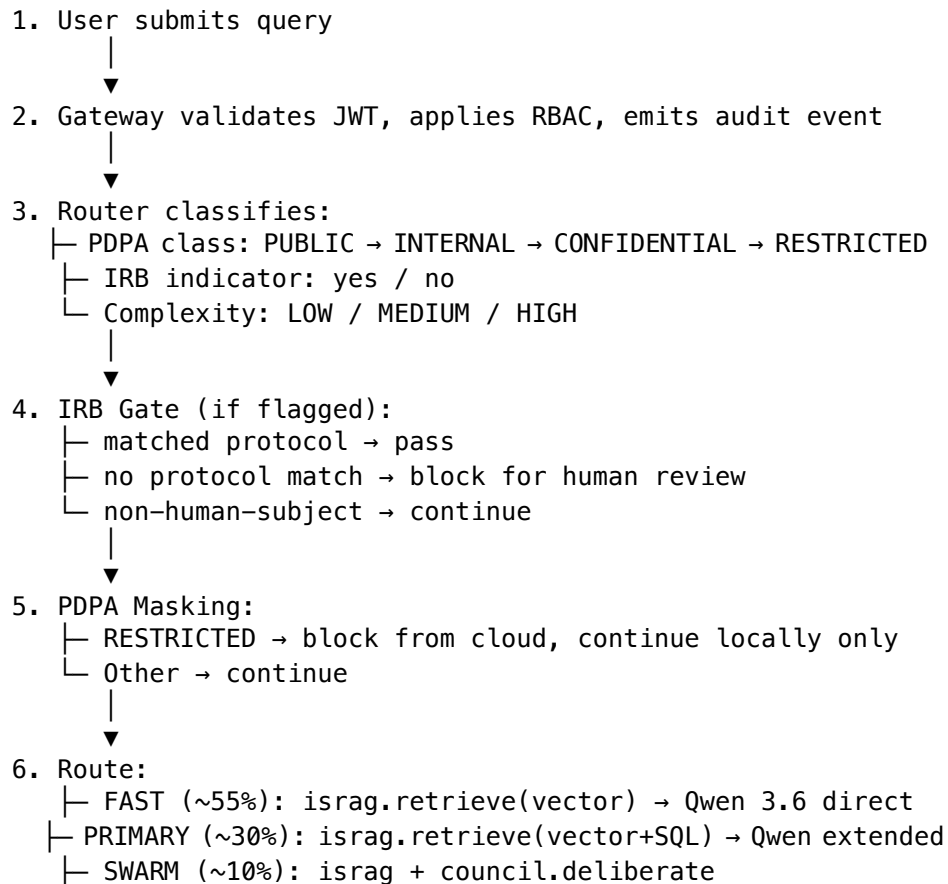
3.2 Package Composition

packages/suluh/ (orchestrator – depends on both leaf packages)



This separation is deliberate. A researcher can `pip install israg` standalone. A developer can `pip install council` standalone. A university IT team installs the whole ecosystem.

3.3 Query Routing Decision Tree



```

└─ ESCALATE (<5%): anonymise → Anthropic → return
    |
    ▼
7. Synthesis + Isnad chain assembly → Response to user

```

3.4 Data Model: Cross-Package Types

IsRAG defines the **canonical data structures** that flow through the ecosystem:

```

KnowledgeEntry {
  content: str
  entry_id: str (hash-based)
  author: str
  author_type: str
  source_type: SourceType (8 types)
  verification_level: VerificationLevel (6 levels)
  cultural_context: CulturalContext | None
  trust_score: float (computed)
  provenance: list[ProvenanceStep]
  verifications: list[VerificationRecord]
  tags: list[str]
  created_at: datetime
}

```

TrustEngine computes 5-factor score:

```

verification_level (30%) + source_credibility (25%) +
verification_quality (20%) + cross_reference (15%) + freshness (10%)

```

SourceType: RESEARCH · INSTITUTIONAL · GOVERNMENT · ACADEMIC · NEWS ·
EXPERT · COMMUNITY · AI_GENERATED · FIELD_DATA

VerificationLevel (0–5):

```

UNVERIFIED → SELF_REPORTED → PEER_REVIEWED →
INSTITUTION_VERIFIED → MULTI_SOURCE_CONFIRMED → FIELD_VALIDATED

```

CulturalFramework (14):

```

GLOBAL · WESTERN · EAST_ASIAN · SOUTHEAST_ASIAN · SOUTH_ASIAN ·
INDONESIAN · MALAYSIAN · THAI · VIETNAMESE · FILIPINO ·
SINGAPOREAN · ISLAMIC · BUDDHIST · INDIGENOUS

```

Council **consumes** IsRAG types:

```

CouncilMember: name, role (chairman|member), model_id, specialty, trust_weight
Opinion: member, response, confidence, timestamp
Review: reviewer, reviewed, rating (1–10), feedback, issues_found
DeliberationResult: final_answer, isnad_chain, consensus_score, dissenting_views
SULUH wraps both:

```



```

QueryRequest: user_id_hash, routing_label, pdpa_classification,
               irb_flagged, query, context
QueryResponse: answer, isnad_chain, deliberation, pdpa_masking_applied,
               irb_gate_decision, confidence, citations, audit_log_entry

```

4. Package Specifications

4.1 IsRAG — Detailed API

Location: packages/israg/src/israg/israg_engine.py (783 lines)

```

class IsRAGEngine:
    def __init__(self, storage_path: Optional[str] = None)
    def ingest(content, author, source_type, author_type="person",
               verification_level=UNVERIFIED, cultural_context=None,
               tags=None, metadata=None, entry_id=None) -> KnowledgeEntry
    def retrieve(query, min_trust_score=0.0, cultural_filter=None,
               source_filter=None, limit=10, include_isnad=False) -> list[RetrievalResult]
    def add_verification(entry_id, verifier, verifier_type, level,
                       rating, feedback="", issues_found=None) -> KnowledgeEntry
    def get_entry(entry_id) -> KnowledgeEntry | None
    def list_entries(source_type=None, min_trust=0.0) -> list[KnowledgeEntry]
    def get_stats() -> dict

```

RetrievalResult:

```

@dataclass
class RetrievalResult:
    content: str
    entry_id: str
    relevance_score: float
    trust_score: float
    confidence_level: str # very_high, high, medium, low, unverified
    isnad_chain: IsnadChain | None
    source_type: SourceType
    cultural_context: CulturalContext | None

    def format_for_display(show_isnad=False) -> str
    def to_dict() -> dict

```

Current limitations to address in Phase 1: - `_calculate_relevance()` uses naive keyword matching — replace with vector similarity (bge-m3 embeddings + pgvector HNSW) - `knowledge_base` is an in-memory dict — replace with persistent store (PostgreSQL + pgvector) - No embedding model integration yet — need sentence-transformers or bge-m3 via HuggingFace

4.2 Dewan Council — Detailed API

Location: packages/council/src/council/council_engine.py (775 lines)

```
class DewanCouncil:
    def __init__(self, api_key=None, members=None, provider="openrouter",
                 verbose=False)
    def deliberate(query, skip_reviews=False, max_reviewers=3) -> DeliberationResult

    # Internal stages
    def _stage1_get_opinions(query) -> list[Opinion]
    def _stage2_cross_review(query, opinions, max_reviewers) -> list[Review]
    def _stage3_synthesize(query, opinions, reviews) -> dict
    def _call_llm(member, prompt) -> str

    # Mock mode: set no API key and council auto-generates plausible mock responses
    # for testing the deliberation protocol without burning tokens.
```

Reference Council (7 members, OpenRouter model IDs):

Role	Name	Model	Specialty
Chairman	Ketua Claude	anthropic/claude-3-opus-4.7	Synthesis, judgment
Member	Anggota GPT	openai/gpt-4o	Deep reasoning
Member	Anggota Gemini	google/gemini-1.5-flash-preview	Data, multimodal
Member	Anggota Kimi	moonshotai/kimi-k2.6	Long context, multi-agent
Member	Anggota Deepseek	deepseek/deepseek-v4-flash	Chain-of-thought
Member	Anggota Qwen	qwen/qwen3.6-plus	Multilingual, Asia
Member	Anggota Nemotron	nvidia/nemotron-3-super	Open model, free tier

Current limitations to address: - Mock mode is helpful but should be extended with structured test data - No streaming support — all deliberation is synchronous - Chairman selection is hardcoded (first member with role="chairman") - No learned chairman selection or dynamic council composition

4.3 SULUH AI — Planned Components

Location: packages/suluh/ (skeleton only)

Component	Path	Function	Phase
Gateway	suluh/gateway/	FastAPI, JWT, RBAC, audit, SSE	Phase 0
Router	suluh/router/	Query classifier, complexity scorer	Phase 0
IRB Gate	suluh/irb/	Deterministic ethics classifier	Phase 0
Hybrid RAG	suluh/rag/	pgvector + Text-to-SQL	Phase 1
PDPA Masking	suluh/pdpa/	PII detection, fail-closed	Phase 1
Docling	suluh/docling/	Document intelligence pipeline	Phase 1
Swarm	suluh/swarm/	LangGraph coordinator	Phase 2
Registry	suluh/registry/	Agent catalogue	Phase 1
Frontend	frontend/	Vue 3 Workbench	Phase 1

Gateway API contract (planned):

```
GET /health
POST /query          - main query endpoint
GET /sessions        - list user sessions
GET /sessions/{id}   - get session with full audit trail
POST /sessions/{id}/stop - stop running session
GET /models          - list available models
GET /agents          - list registered agents
GET /audit            - query audit log (admin)
```

Router contract (planned):

```
class QueryRouter:
    def classify(query: str, user_context: UserContext) -> RouteDecision:
        """
        Returns:
            RouteDecision {
                pdpa_class: PDPAClass (PUBLIC|INTERNAL|CONFIDENTIAL|RESTRICTED)
                irb_flagged: bool
            }
```

```

        complexity: ComplexityClass (LOW|MEDIUM|HIGH)
        route: Route (FAST|PRIMARY|SWARM|ESCALATE)
        confidence_threshold: float
    }
}

```

Swarm Coordinator — 6-step LangGraph pipeline (planned):

1. query_analyzer — understand intent, scope, domain
2. discovery — identify relevant agents, tools, sources
3. decompose — break into subtasks, assign to agents
4. execute — run agent tasks in parallel subgraphs
5. consensus — aggregate results, resolve conflicts
6. irb_gate — ethics check before returning to user

5. Compliance Architecture

5.1 PDPA Data Classification

Level	Examples	Cloud escalation	Retention
PUBLIC	Published papers, anonymised stats, open datasets	Permitted	Indefinite
INTERNAL	Research aggregates, non-PII project data	Permitted (anonymised)	7 years
CONFIDENTIAL	Researcher profiles, grant apps, pre-publication manuscripts	Prohibited	7 years
RESTRICTED	Participant IDs, interview transcripts, clinical data, IRB records	Prohibited	10 years

Enforcement points: - `suluh.pdpa.classify()` — every document, query, and tool output is classified - `suluh.pdpa.mask()` — CONFIDENTIAL and RESTRICTED data masked before processing - RESTRICTED queries never

leave the intranet regardless of any other condition - **Fail-closed**: if the masker errors, the query is blocked, not passed through

5.2 IRB Research Ethics Controls

The IRB gate runs on **every query** before routing. It uses a deterministic decision layer — even when the underlying classifier is LLM-based, the decision rule is a Python function.

Five-question pre-screen: 1. Human participants involved? (directly or via their data) 2. Identifiable personal information? 3. Sensitive categories (health, religion, ethnicity, political opinion)? 4. Vulnerable populations? 5. Has an IRB protocol been approved for this research type?

Any “yes” on (1), (2), or (3) without an approved protocol □ **IRB gate blocks**.

Five research types for Phase 0 classifier: - Survey research (questionnaires, interviews) - Clinical / medical research (patient data) - Educational research (student data, classroom observation) - Indigenous community research (adat, customary knowledge) - Vulnerable population research (children, elderly, prisoners)

5.3 Audit Log (Immutable)

The `ai_audit_log` PostgreSQL table is append-only — UPDATE and DELETE are rejected by trigger. All events retained 7–10 years per PDPA.

Key events: AI query submitted, SQL executed, agent swarm invoked, Anthropic escalation, PDPA PII detection, IRB gate decision, document lifecycle change, routing config change, failed authentication.

5.4 Network Perimeter

- The EXO cluster, AI replica, MinIO, Redis, and LangFuse are on the institution’s **intranet only**
- Edge DGX Spark nodes are on isolated faculty VLANs with firewall rules
- FastAPI gateway is accessible from campus network and VPN — **never internet-facing**
- Only outbound connection permitted: HTTPS to `api.anthropic.com` from quality router, and only when all conditions met
- All intra-service communication: TLS 1.3

6. Design Principles (7 Immutable)

These are **constraints, not preferences**. Any proposal that violates them requires a formal Steering Committee design amendment.

1. **Sovereign by default** — all data, state, model weights stay inside Malaysian jurisdiction and institutional network perimeter.
2. **Read-only by design** — agents recommend and synthesise; no write to production systems. Write capability requires Steering Committee approval.
3. **Hybrid retrieval, not pure RAG** — vector search for documents + SQL generation for structured data + deterministic rules for compliance. Pure semantic RAG will produce incorrect research outputs.
4. **Document intelligence is mandatory** — Docling-grade OCR, table extraction, figure extraction from day one, not a Phase 2 enhancement.
5. **Auditability over novelty** — every agent action is explainable, attributable to source agent + model version, and logged to immutable audit trail.
6. **Evaluation-driven deployment** — no model, config, prompt, or workflow change reaches production without passing the golden set regression gate.
7. **Model-agnostic architecture** — swap the inference model by updating one config value. No code changes in the orchestrator or retrieval layers.

Each principle is **operationalised in code** — it is enforced at the application level, not at the configuration level. A misconfigured environment variable cannot bypass a principle.

7. Development Roadmap

7.1 Phase 0: Foundation (Months 1–2)

Validation sprint. Answer three questions before committing to full build:

Task	Priority	Deliverable
SSO bridge	P0	UPSI SSO □ JWT validated in FastAPI skeleton
Schema + FastAPI skeleton	P0	schema.yaml merged, gateway passing health check
Bilingual model benchmark	P0	Qwen vs Gemma vs Nemotron on 50 golden-set entries
IRB gate prototype	P0	Keyword classifier for 5 research types, <500ms latency

Task	Priority	Deliverable
Golden set v0.1	P0	50 Malay/English research queries with ground truth
Git initialisation	P0	Repository under version control

Phase 0 runs on a single developer machine — a Mac (M-series, 64 GB) or single DGX Spark. The cluster is not required to start. All Tier 1–3 components are stubbed. Single SQLite or local Postgres for Tier 4.

Exit gate: SSO validated, schema merged, FastAPI skeleton passing, benchmark report accepted, IRB prototype validated, golden set at 50 entries.

7.2 Phase 1: Core Capabilities (Months 3–6)

Component	Deliverable
IsRAG v0.2	Vector search (bge-m3 + pgvector), persistent storage, embedding pipeline
Dewan Council v0.2	Streaming support, async deliberation
Hybrid RAG	pgvector HNSW + Text-to-SQL with 5-stage sqlglot sandbox
Text-to-SQL	Table allowlist, statement-type allowlist, row-limit, query timeout
Docling pipeline	OCR, table, figure extraction; low-confidence flagging for HITL review
PDPA masking	presidio-based PII detection, bilingual Malay/English
IRB gate v1	Fine-tuned classifier on 200 labelled queries
Agent Registry	Agent catalogue, capability manifests, signed agent deployment
Chat Workbench	Vue 3 UI: chat, file tree, editor, terminal, session history
LangFuse	Self-hosted observability, spans at every layer
Golden set v0.2	200 entries across 5 domains

Exit gate: All Tier 1–3 components on staging, Month 6 metrics: accuracy 80%, adoption 25%, latency < 4s, PDPA 100%.

7.3 Phase 2: Agentic Workflows (Months 7–10)

Component	Deliverable
Multi-Agent Swarm	LangGraph v1.0, 6-step pipeline, max 15 steps/workflow
Research Experiment Automation	Agentic workflow for designing, running, and reporting experiments
IRB Compliance Auditing	Automated IRB workflow audit, 10% random sample review
Regression CI	Golden set gates every deployment; PR blocks on failure
IRB Audit Dashboard	Admin console showing gate decisions, flagged queries, protocol expiry
Golden set v1.0	200 entries, expanded to cover consortium domains

Exit gate: 3 workflows live, Month 10 metrics: accuracy 88%, adoption 55%, latency < 3s, Text-to-SQL 85%, swarm success 80%, IRB gate 95%.

7.4 Phase 3: Full Production (Months 11–12)

Activity	Target
Production rollout	75% adoption across research staff and graduate students
Admin Console	Monitoring, user management, audit review
Consortium API	Federation spec documented and tested
Post-go-live monitoring	LangFuse dashboards, weekly IRB audit samples
Annual compliance attestation	Signed by IRB Chair, Legal/Compliance lead, Lead Architect

Exit gate: 75% adoption, consortium API tested, all metrics at target: accuracy 93%, swarm success 90%, latency < 2.5s, IRB gate 98%, Anthropic escalation < 5%.

8. Golden Set & Evaluation Plan

8.1 Golden Set Specification

- **50 entries (Phase 0)** □ **200 entries (Phase 1)** □ **200+ entries (Phase 2+)**
- **5 domains:** agriculture, Islamic finance, indigenous land rights, public health, digital governance
- **Bilingual:** Malay and English
- **Authored by:** 30+ researchers from consortium institutions
- **Licence:** CC-BY 4.0 with author consent; no PII
- **Format:** JSONL with query, golden_answer, domain, language, difficulty, requires_deliberation, irb_flagged

8.2 Evaluation Metrics

Metric	Method	Target
Answer accuracy	BERTScore vs golden answer	93% (Month 12)
Trust calibration	Pre/post Likert survey (100+ users)	Effect size > 0.5
Cultural appropriateness	Expert review, 3-point ordinal	90% appropriate
Hallucination rate	Claimed citations that don't exist or are misattributed	< 5%
Provenance completeness	Does answer include full chain?	100%
Text-to-SQL correctness	Execution match	92%
Swarm task success	LangFuse trace review	90%
PDPA compliance	Audit log review	100%
IRB gate accuracy	IRB officer review	98%

8.3 Model Benchmark (Phase 0)

Candidate	Size	Provider	Notes
Qwen 3.6 35B-A3B	35B (3B active)	Alibaba	Primary candidate, MoE, bilingual
Gemma 4-31B	31B	Google	Strong multilingual

Candidate	Size	Provider	Notes
Nemotron-3 Super	?	NVIDIA	Strong tool-calling, open model

Benchmark on 50 golden set entries, measuring BERTScore, latency, and token efficiency. Winner becomes the primary model; swap paths maintained for all three.

9. Hardware & Infrastructure

9.1 Target Cluster (Phase 1+)

Node	Hardware	RAM	Role
3× DGX Spark	NVIDIA Founders Edition	128 GB each	Edge nodes, data-locality
2× Mac Studio	M3 Ultra	256 GB each	Core inference nodes
Total	5 nodes	1,280 GB pooled	

- **Interconnect:** ConnectX-7 200GbE fabric
- **Disaggregation:** EXO — edge prefill, core decode split
- **All on-premise:** No cloud inference for primary models

9.2 Phase 0 Dev Environment

- Single Mac (M-series, 64 GB+) or single DGX Spark
- EXO configured for single-node inference
- SQLite or local Postgres for Tier 4
- LangFuse on localhost:3000
- Vue 3 Workbench via `npm run dev`

9.3 Software Stack

All open-source. RM 0 licensing:

Layer	Technology
Models	Qwen 3.6, Nemotron-3 Super, Gemma 4, bge-m3
Orchestration	LangGraph v1.0

Layer	Technology
Gateway	FastAPI (Python)
Frontend	Vue 3 + Tailwind
Vector DB	pgvector (HNSW index)
Document	IBM Docling
Observability	LangFuse (self-hosted)
Cache/State	Redis
Disaggregation	EXO
CDC	Debezium / pg_logical
Object Store	MinIO

10. Team & Resources

Role	Allocation	Responsibilities
Putra Nasution — Lead Architect & PI	1.0 FTE · 12 months	Programme design, all architecture and code, paper authorship
Engineering team (remote)	0.5 FTE · 12 months	Frontend (Vue 3 Workbench), testing, documentation, LangFuse instrumentation
Dr. Suzani Mohamad Samuri — Oversight Director / Co-PI	50% · 12 months	Architecture oversight, IRB gate validation, paper co-authorship, stakeholder engagement
UPSI DBA Team	20% Phase 0–1	PostgreSQL replication, AI read replica provisioning
UPSI Research Community	20% Phase 0	Golden set authorship, query validation, feedback
UPSI IRB / Ethics Committee	Milestone-based	Ethics workflow review, IRB gate validation
UPSI Legal & Compliance	Milestone-based	PDPA compliance sign-off, audit engagement

Budget: RM 285,000 (~USD 60,000)

Category	RM
Hardware (3x DGX Spark + 2x Mac Studio + networking)	175,000
Software licensing (open source)	0
Cloud (Anthropic API, <5% queries)	5,000
Personnel (0.5 FTE x 12 months)	60,000
Training & change management	15,000
Travel & publication	20,000
Contingency (~3.5%)	10,000

PI effort is in-kind. OpenRouter costs for Dewan Council research are covered by existing research credits.

11. Risk Register

Risk	Likelihood	Impact	Mitigation
SSO integration complexity	Medium	High	Phase 0 critical path, Week 1; Phase 1 cannot begin until validated
Model underperforms on bilingual	Medium	High	Phase 0 benchmark: Qwen vs Gemma vs Nemotron; swap paths maintained
Multi-agent consensus produces misleading outputs	Medium	High	Confidence-gated display, IRB gate, regression golden set gates
IRB gate adds latency	Medium	Medium	Async pre-screening, fast-path for non-human-subject; <500ms target

Risk	Likelihood	Impact	Mitigation
PI bandwidth split across programmes	High	High	Engineering team absorbs implementation; PI focuses on architecture + papers
Hardware procurement delay	Medium	High	Phase 0 runs on a single Mac if cluster slips; 6-week fast-track procurement
Academic adoption lower than forecast	Medium	Medium	Early adopter programme, researcher workshops, embedded support
Open-source model quality regression	Low	High	Model swap paths maintained; Claude API escalation as safety net

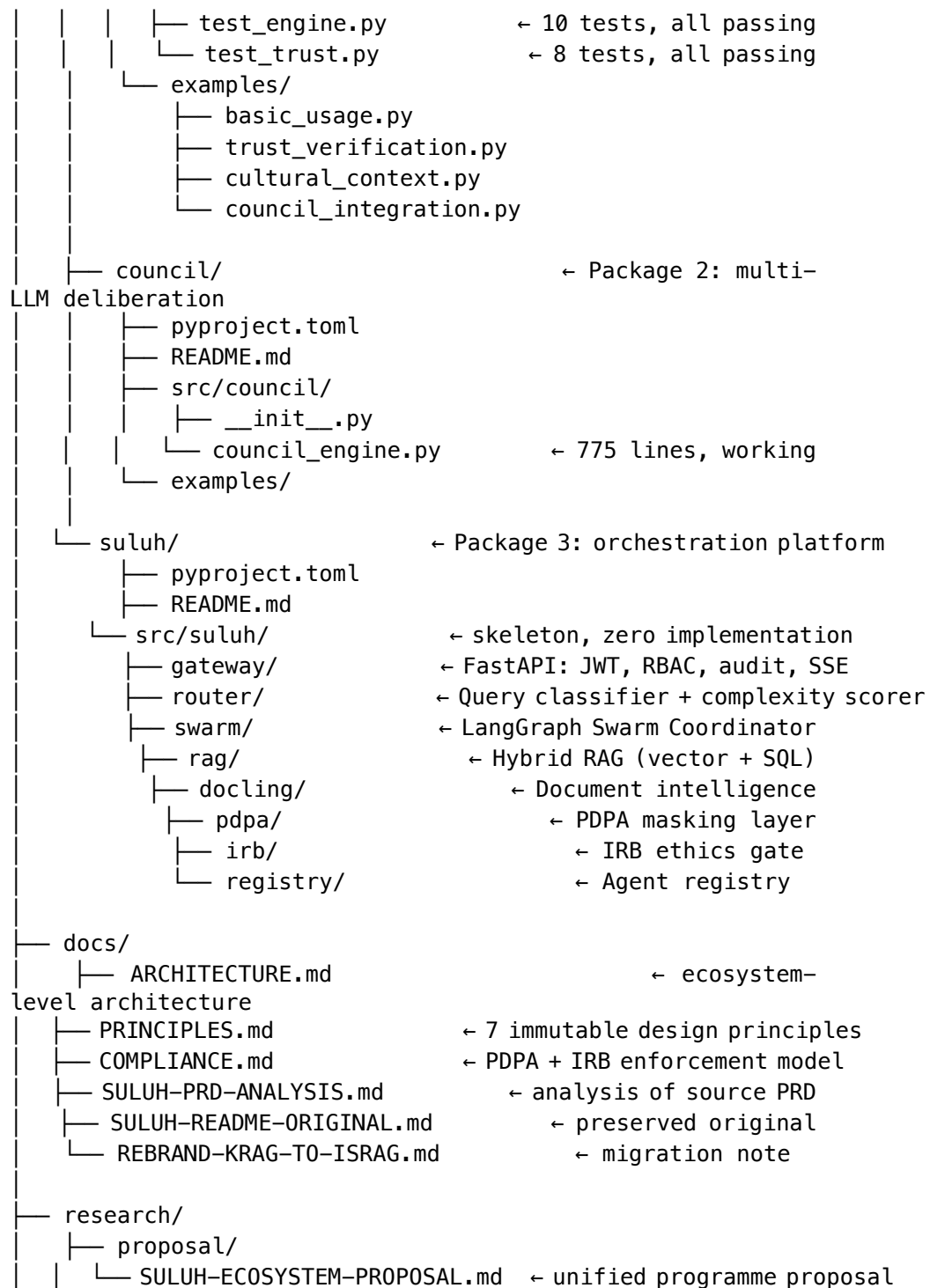
12. Repository Structure

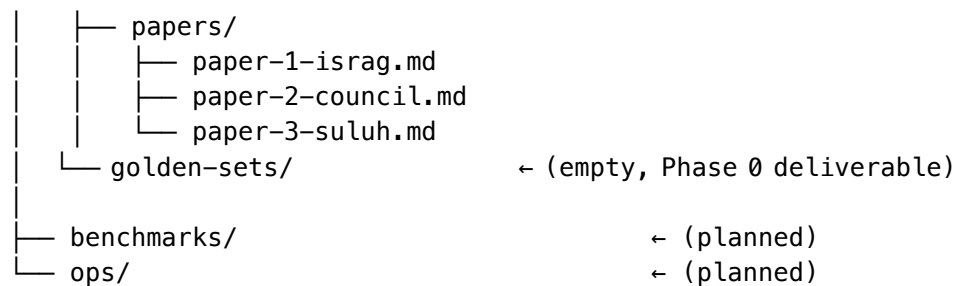
```

suluh-ecosystem/
├── README.md
├── pyproject.toml
├── LICENSE
├── packages/
│   └── israg/
aware RAG
├── pyproject.toml
├── README.md
├── src/israg/
│   ├── __init__.py
│   └── israg_engine.py
└── tests/

```

← private monorepo
 ← ecosystem-level overview
 ← workspace root
 ← Apache 2.0
 ← Package 1: provenance-
 ← 783 lines, working





13. Developer Quick Start

```

cd suluh-ecosystem
pip install -e "packages/israg[dev]"
pip install -e "packages/council[dev]"
pip install -e "packages/suluh[dev]"

# Test IsRAG
python -m pytest packages/israg/tests/ -v

# Run IsRAG engine
python -c "
from israg import IsRAGEngine, SourceType
e = IsRAGEngine()
e.ingest(content='ASEAN has 670M people', author='UN', source_type=SourceType.GOVERNMENT)
print(e.retrieve('ASEAN population'))
"

# Run Dewan Council (mock mode – no API key needed)
python -c "
from council import DewanCouncil
c = DewanCouncil()
result = c.deliberate('What is the best AI strategy for ASEAN higher education?')
print(result.final_answer)
"

```

14. Open Questions for OpenScience

These are the questions OpenScience should be aware of as it researches the project:

1. **Naming:** Is “SULUH” the final ecosystem name, or should we reserve space for a model-level branding (e.g., SULUH-EDU)?

2. **Training data (“100+ years of UPSI data”)**: What is the actual composition? Academic content (theses, journals), institutional admin (PDPA-sensitive), Malay-language education corpus, or all three? The training pipeline, PDPA posture, and licensing all depend on this.
3. **Socratic AI features**: What does “Socratic” mean pedagogically? Dialogue-based guided inquiry, adaptive tutoring (Khanmigo-style), or lesson plan generation? The SFT dataset design and eval methodology change significantly.
4. **Model training strategy**: QLoRA (fast, 95% of full FT quality), continued pretraining (CPT) on UPSI corpus, or full fine-tune? QLoRA is the default recommendation for Phase 0 with escalation to CPT if results don’t meet the accuracy bar.
5. **Consortium expansion**: Phase 3 federation design — is it a product line, a consulting offering, or a hosted inference service? Three very different operating models.
6. **Open-source licence**: The codebase is Apache 2.0. The model weights will need a separate licence decision (Apache-style, Llama-style restrictions, or open-weights-not-open-source).
7. **Phase 2+ budget**: The RM 285K covers Phase 0–3 (12 months). Phase 2 expansion (public inference, consortium launch) may need RM 5M — timeline and fundraising vehicle TBD.

15. Relationship to Other Projects

~/Documents/PROJECTS/
├─ suluh-ecosystem/ ← THIS REPOSITORY (consolidated, private)
├─ krag/ ← legacy: superseded by packages/israg/
├─ dewan-council/ ← legacy: superseded by packages/council/
├─ suluh-ai/ ← legacy: superseded by packages/suluh/
└─ INTELIGENSIA-English/ ← book; SULUH ecosystem referenced as Chapter 12

End of PRD. This document should be treated as the canonical product specification for the SULUH Ecosystem. The code, the architecture docs, and the papers all defer to this file. Update it when the architecture or scope changes.

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